

UNIT -I

Business Intelligence

Business Intelligence:

Effective and timely decisions, Data, information and knowledge, the role of mathematical models, Business intelligence architectures, Ethics and business intelligence

Business Intelligence:

It means using tools and methods to analyze data and turn it into useful information. This helps in making better and more informed decisions.

Key features:

a. Data collection :

- Gathering data from various sources like sales systems, websites or customers feedback.
- Example: A store collects data on daily sales and customer purchases.

b. Data Storage :

- Storing raw data in central location, such as a data warehouse.
- Example : Sales and inventory data are saved in a single database for easy access.

c. Data Analysis :

- Analysing the data to find patterns, trends and useful information.
- Example : Analysing sales trends to find which product sells the most.

d. Visualization :

- Presenting data in charts , graphs or dashboards for easy understanding.
- Example : A graph showing monthly sales growth to managers.

e. Decision-Making Support :

- Using understandings to make informed decisions.
- Example: A company increases stock of a popular product after analysing sales trends.

In Short, BI helps businesses use data to improve decision making, identifies business opportunities and problem areas, enhance efficiency, helps in forecasting future trends. (To make smarter decisions and grow effectively)

Effective and timely decisions:

Why Effective and Timely Decisions Are Important

Let's understand with an example:

Case Study:

Example 1.1 – Mobile Phone Industry:

- **Mobile Phone Industry**

1. A marketing manager notices many customers are leaving the company for competitors (this is called customer churn).
 2. Customer churn is a big problem for companies in service industries because it lowers profits.
- **The Problem:**
 1. The manager has a budget to run a campaign to retain 2,000 customers out of 2 million.
 2. The challenge is to choose which 2,000 customers to target for maximum impact.
 - **The Solution:**
 1. Predict which customers are most likely to leave using data analysis.
 2. Focus on customers with a high chance of leaving but who are also valuable to the company.
 - **The Importance of Tools:**
 1. Advanced tools like mathematical models and data mining are needed to predict churn probabilities accurately.
 2. Without these tools, it would be very hard to identify the right customers and make effective decisions.
 - **Outcome:**
 1. By targeting the right customers, the company can reduce churn and keep valuable customers, improving overall business success.

Example 1.2 – Logistics Planning

Logistics means planning and managing the movement of goods or resources from one place to another.

Example: Delivering packages from an online store to customers' homes.

1. The Problem:

- A logistics manager needs to create a plan for production and logistics over the next year.
 - The plan includes:
 - Assigning demand from different markets to production sites.
 - Getting raw materials from suppliers.
 - Planning production at factories.
 - Distributing finished products to customers.
- 2. The Challenge:**
- The company may have many factories, suppliers, and thousands of products to manage.
 - The process is very complex and involves decisions for a year, broken down into weeks.
- 3. Why Intuition is Not Enough:**
- Many people make decisions based on experience, knowledge, or simple methods.
 - This approach doesn't work well in today's fast-changing business environment.
- 4. The Solution:**
- Advanced tools, like optimization models and mathematical methods, are needed to handle complex logistics problems.
 - These tools help make smarter and more effective decisions in a dynamic and competitive world.
- 5. Outcome:**
- Using analytical tools leads to better plans and helps the company adapt to changes quickly.

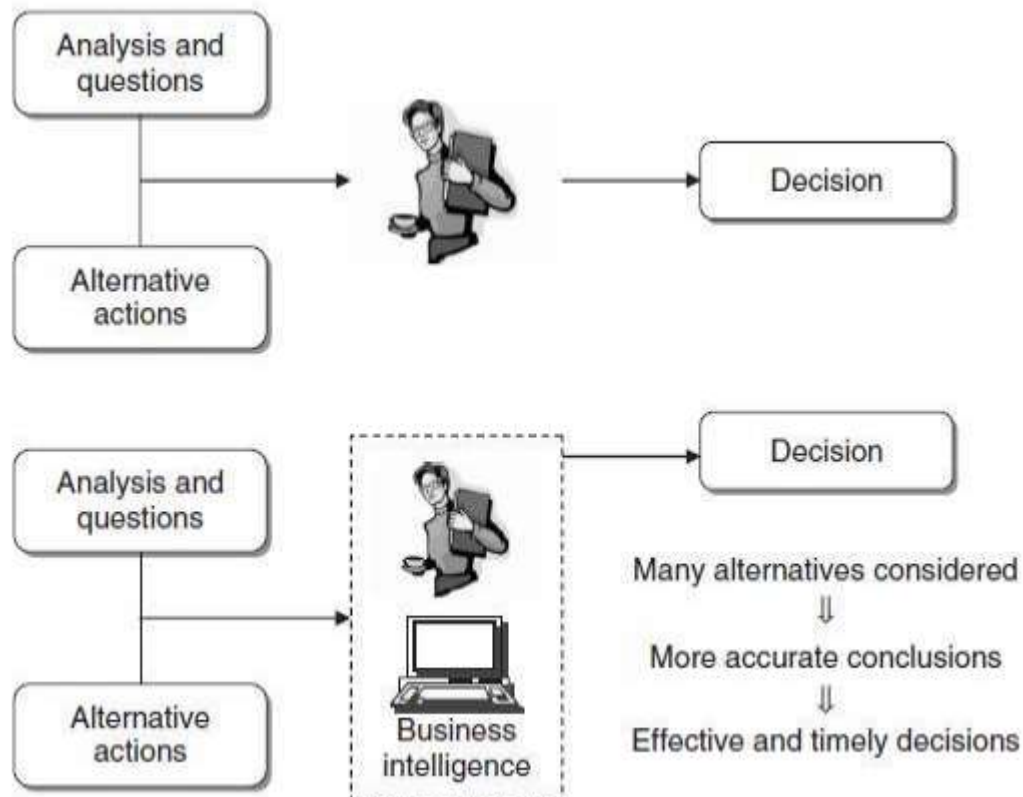
Effective Decisions

- 1. Better Information and Knowledge:**
 - Using analytical methods provides more reliable and accurate information for decision-making.
- 2. Improved Decisions and Plans:**
 - Decision-makers can create better action plans to achieve their goals effectively.
- 3. Clear Evaluation of Choices:**

- Analytical methods help clearly define criteria for comparing different options.
4. **Deeper Understanding:**
- The process encourages decision-makers to think deeply, understand the problem better, and improve their decision-making logic.

Timely Decisions

1. **Fast-Changing Environment:**
- Businesses face tough competition and constant changes in the market.
2. **Need for Quick Actions:**
- Companies must respond quickly to competitors' moves and changing market conditions.
3. **Critical for Success:**
- Acting on time is essential for a company's success and survival.



Data, information and knowledge:

1. Data

- **Definition:**

Data refers to raw, unprocessed facts or figures related to individual entities or transactions between entities. It is usually structured but lacks context or meaning by itself.

- **Examples in Business:**

- Primary entities for a retailer:
 - **Customers:** Name, contact details, loyalty card ID.
 - **Points of Sale (Stores):** Location, store ID, operational hours.
 - **Items (Products):** Product name, price, stock level.
- Transactions: Sales receipts showing details like items purchased, quantities, prices, and dates.

- **Key Idea:**

Data serves as the foundational building block for creating meaningful insights. Without data, further processing for information and knowledge is not possible.

2. Information

- **Definition:**

Information is created by processing, organizing, and analyzing raw data to give it meaning in a specific context. Information is actionable and useful for decision-making.

- **Examples in Business:**

- The sales manager analyzes raw sales data and finds meaningful patterns, such as:
 - The percentage of weekly sales receipts exceeding €100.
 - A list of loyalty card customers whose spending dropped by over 50% in the last three months.

- **How It Is Used:**

- Information allows businesses to identify trends, monitor performance, and detect issues (e.g., declining customer spending).

- **Key Idea:**

Information transforms raw data into meaningful insights, tailored for a specific purpose or problem.

3. Knowledge

- **Definition:**

Knowledge is the application of information, enhanced by human experience, expertise, and judgment, to solve complex problems or make informed decisions.

- **Examples in Business:**

- Based on information about reduced spending by customers in a specific area, the business:
 - Identifies the cause (e.g., a competitor opened a store nearby).
 - Takes action (e.g., introduces free home delivery for customers in that area).

- **Key Aspects of Knowledge Creation:**

- **Understanding Cause-and-Effect Relationships:** Helps in identifying why certain trends or problems are happening.
- **Decision-Making:** Helps businesses act on insights to solve problems or seize opportunities.

4. How Knowledge is extracted from Data

- **Passive Approach:**

- Decision-makers suggest specific criteria for analysis (e.g., find customers with reduced spending).
- Analysts use these criteria to extract useful knowledge from data.

- **Active Approach:**

- Advanced methods like mathematical models, inductive learning (e.g., machine learning), or optimization algorithms are used to uncover hidden patterns or generate recommendations.

5. Knowledge Management

- **Definition:**

Knowledge management involves integrating technology and decision-making processes to efficiently turn data into actionable knowledge.

- **Purpose:**

- To help businesses make better decisions and solve problems quickly and effectively.
- **How It Works:**
 - Combines tools (e.g., data analysis software, databases) with processes (e.g., decision-making frameworks) to support knowledge workers.
- **Outcome:**
 - Ensures that raw data evolves into meaningful insights and actionable strategies to improve performance.

Summary

- **Data** is the raw foundation.
- **Information** organizes and processes data into meaningful insights.
- **Knowledge** applies this information, combined with experience, to solve problems or make decisions.
- Knowledge extraction can be **passive** (based on predefined criteria) or **active** (using advanced tools).
- **Knowledge management** integrates technology and processes to support informed, effective decision-making.

This hierarchy ensures businesses can move from simply collecting data to driving smart, informed actions that enhance their success.

Differentiate between Data, Information and Knowledge:

Aspect	Data	Information	Knowledge
Definition	Raw, unprocessed facts or figures without meaning.	Processed and organized data that provides meaning in a specific context.	Insights or actionable understanding derived from information, enhanced by experience and expertise.
Nature	Basic and unorganized.	Structured and organized.	Applied and contextualized.
Purpose	To serve as the foundation for processing and analysis.	To provide meaningful insights for understanding a specific situation.	To solve problems or make decisions based on deeper understanding.
Example	A list of all transactions, e.g., sales receipts, customer details, and product IDs.	Summary reports, such as "10% of customers spent more than €100 this week" or "Customer X reduced spend  by 50% in the last three months."	A decision to offer free home delivery to customers in an area where competitors have opened a new store, based on insights from reduced customer spending.
Processing Required	No processing required (raw data).	Requires analysis, aggregation, or summarization to extract meaningful patterns.	Requires combining information with experience, expertise, and reasoning to draw actionable insights.
Output	Numbers, text, or raw files without immediate use.	Graphs, tables, or reports with specific meaning for a domain.	Action plans, strategies, or solutions for solving complex problems.
Use in BI	Acts as input for Business Intelligence systems to process and analyze.	Provides insights for specific business scenarios, such as performance tracking or trend analysis.	Enables businesses to make informed decisions and take actions to address issues, capitalize on opportunities, or improve overall efficiency.
Tools/Methods	Stored in databases (e.g., SQL databases).	Extracted using query tools, reporting systems, or data visualization tools. 	Created through decision-making processes, knowledge management systems, or advanced analytics tools like machine learning and optimization.
Dependency	Independent (no prior input needed).	Depends on data for its generation.	Depends on information and the decision-maker's expertise for effective application.
Value in BI	Low, as raw data lacks context and meaning.	Medium, as information provides actionable insights but doesn't address how to act on them.	High, as knowledge combines insights with strategies for achieving business goals.

The Role of Mathematical Model:

1. Purpose:

A business intelligence system helps decision-makers use data to make better decisions by applying mathematical models and algorithms.

Example:

A store manager uses a business intelligence system to understand which products are selling the most, based on sales data, to decide what products to restock.

2. Simple Tasks:

For basic tasks, the system may calculate totals and percentages and show them in simple charts (like bar or line graphs).

Example:

The system calculates that 20% of products sold in the last month were from a new collection, and it shows this in a bar chart for easy understanding.

3. Complex Tasks:

For deeper analysis, more advanced models are used, such as optimizing sales predictions or finding patterns in customer behavior.

Example:

The system uses a complex model to predict which items will sell the most in the upcoming season based on past trends.

Steps in Business Intelligence Analysis:

1. Identify Goals:

First, you need to figure out what you want to achieve with the analysis.

Example:

A restaurant manager wants to find out which menu items are the most popular so they can create a new promotion.

2. Define Performance Indicators:

Decide on the metrics or measures to judge success.

Example:

The manager might use "number of orders per item" or "profit per dish" to decide which items are successful.

3. Develop Math Models:

Create models that explain how different factors (like price or season) affect the results.

Example:

A model is built to show how a price change in a popular dish affects the number of orders it gets.

4. What-If Analysis:

Test how changes in factors (like discounts, stock, or seasonality) affect performance.

Example:

The manager runs a "what-if" analysis to see how giving a 10% discount on the most popular dish would increase sales and profits.

Advantages of Mathematical Models:**1. Better Focus:**

Building a mathematical model helps decision-makers concentrate on the most important parts of the problem, leading to better understanding.

Example:

By creating a model, the restaurant manager understands that the price of a dish is the most significant factor affecting sales.

2. Easier Knowledge Sharing:

The knowledge gained from the model can be easily shared with others in the company, helping everyone make better decisions in the future.

Example:

The manager shares the model with the staff, and they all understand how changing the prices of some dishes could boost sales.

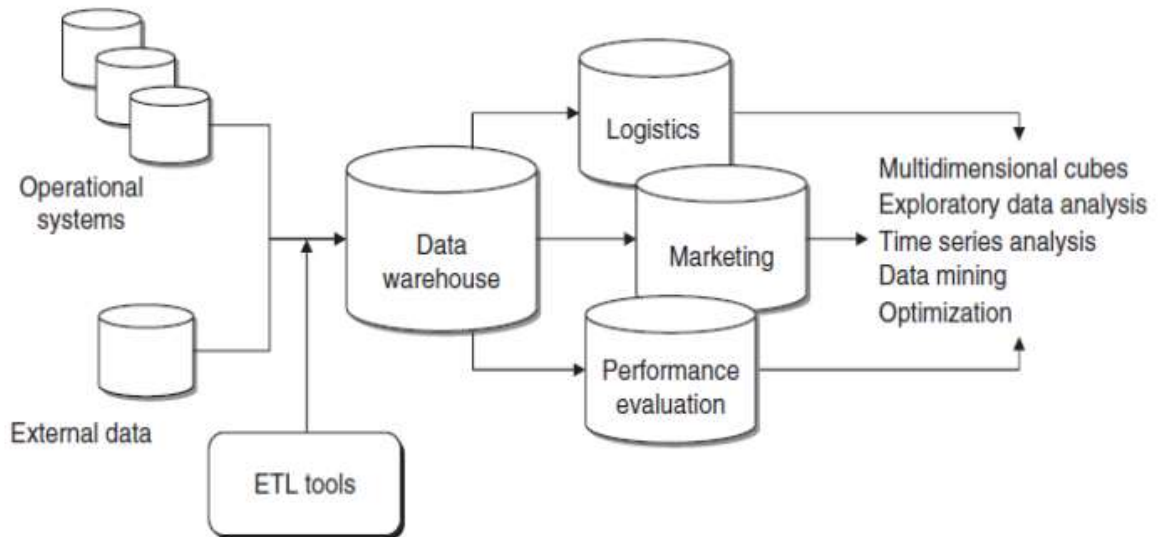
3. Flexibility:

A model made for one situation can often be used in other similar situations.

Example:

The model that worked for the restaurant's pricing decisions can also be applied to a new menu in another branch of the same restaurant chain.

Business intelligence architectures:



1. Data Sources:

The first step is to collect and combine data from different sources, such as operational systems and unstructured data like emails or data from outside providers.

Example:

A company gathers sales data from its store system, customer feedback from emails, and external data from suppliers.

2. **Data Warehouses and Data Marts:**

Data is stored in special databases (data warehouses or marts) after being cleaned and processed using ETL (Extract, Transform, Load) tools. This makes the data ready for analysis.

Example:

The sales data and supplier information are organized and stored in a database for easy access and analysis.

○ **Business Intelligence Methodologies:**

Once the data is stored, it is used for different types of analysis to help decision-makers. These analyses include:

1. **Multidimensional cube analysis**
2. **Exploratory data analysis**
3. **Time series analysis**
4. **Inductive learning models** for data mining
5. **Optimization models**

Example:

A sales manager uses the data to understand how different products perform over time, helping decide what to focus on in the next quarter.

3. **Data Exploration:**

This is the process of searching through data using tools like reports and statistical methods. The goal is to confirm or test hypotheses made by decision-makers.

Example:

A sales manager notices a drop in revenue in a specific area and uses tools to check if this is true and what might be causing it.

4. **Data Mining:**

Data mining helps find hidden patterns or trends in large data sets. It uses techniques like pattern recognition and machine learning to discover new

insights.

Example:

A company uses data mining to identify customer buying habits and predict future purchases.

5. **Optimization:**

This level uses mathematical models to find the best solution from a set of possible actions, often among many options.

Example:

A business uses optimization models to decide the best pricing strategy to maximize profits while staying competitive.

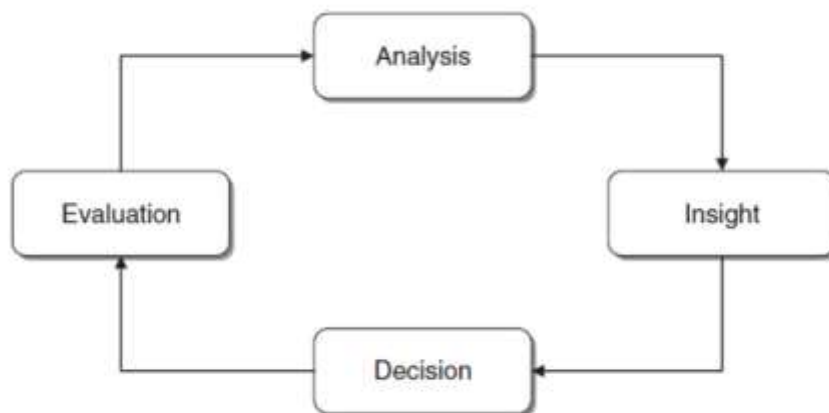
6. **Decisions:**

The final stage is the decision-making process, where the best action is chosen based on all the analysis. Although BI helps with recommendations, the final decision is made by the person in charge.

Example:

After analyzing the data, a manager decides on a marketing strategy, but also considers feedback from the team to adjust the plan.

Cycle of a business intelligence analysis :



1. **Analysis:**

In this phase, the problem is clearly defined. Decision-makers think about what's important and identify the key factors affecting the problem. Business intelligence tools help quickly explore different ways to investigate the problem.

Example:

A manager notices sales are dropping and uses BI tools to ask questions like "What products are underperforming?" or "Is this happening in specific areas?"

2. **Insight:**

In this phase, decision-makers understand the problem better, often by finding the cause. The information from the analysis phase is turned into knowledge. This could come from the decision-maker's experience or through learning models that analyze the data.

Example:

The manager realizes that a new competitor has opened a store nearby, causing the drop in sales. This insight helps to understand the root cause.

3. **Decision:**

After gaining insights, this phase involves making decisions and taking action based on the knowledge gathered.

Example:

Based on the insight, the manager decides to launch a marketing campaign to attract customers back.

4. **Evaluation:**

The final phase is about measuring the results and evaluating performance. It's important to look at various metrics, not just financial, to see how well the decision worked.

Example:

After the marketing campaign, the manager tracks the sales increase, customer feedback, and other performance indicators to evaluate if the decision improved the situation.

Enabling factors in business intelligence projects:

1. Technologies:

- **Hardware and Software:** Advances in technology have made business intelligence systems possible. Over the last 20 years, computer processors have become faster and cheaper. This helps run complex algorithms used in analyzing data.

Example: Microprocessors have improved so much that companies can now use advanced algorithms to predict customer behavior quickly.

- **Data Storage:** The ability to store large amounts of data has increased, and prices have dropped, allowing organizations to store terabytes of data for business analysis.

Example: A company can now store and analyze all their customer transaction data without worrying about space or cost.

- **Network Connectivity:** Technology like Extranets and Intranets helps share information across the company, making business intelligence systems more effective.

Example: Employees across different departments can access and share insights from the business intelligence system instantly.

- **Integration:** Hardware and software from different suppliers or developed internally can easily work together, improving the performance of business intelligence tools.

Example: A company can combine tools from different vendors to analyze their sales data in one system.

2. Analytics:

- **Mathematical Models and Analytical Methods:** These are used to improve data and turn it into useful knowledge. Simple data visualization helps, but advanced models are needed for deeper analysis.

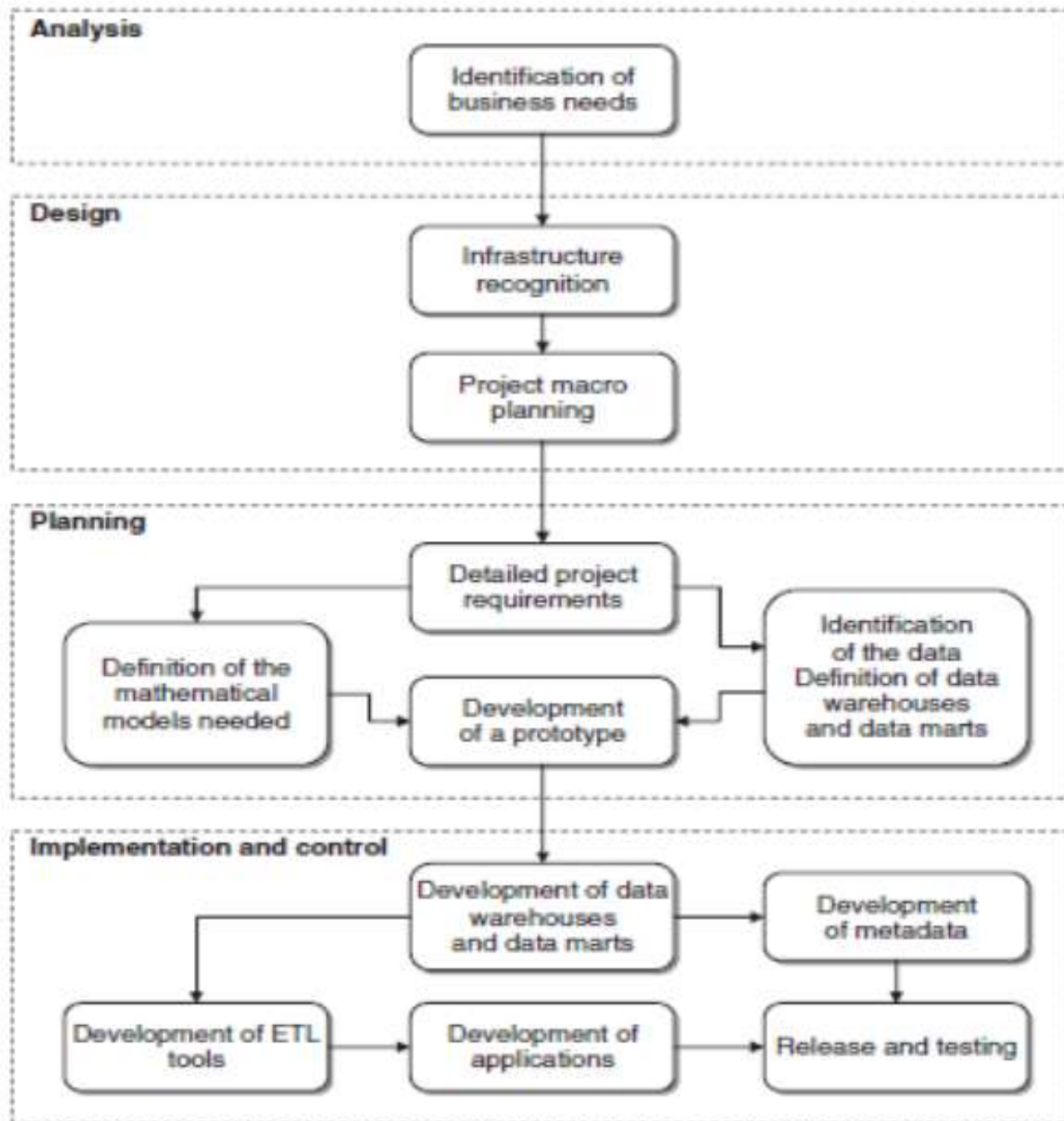
Example: Showing a chart of sales is helpful, but using a mathematical model to predict future sales trends is more useful for decision-making.

3. **Human Resources:**

- **People and Skills:** The success of business intelligence systems also depends on the skills of the people who use them. Knowledge workers (employees who analyze and use data) are key to turning data into actionable insights.

Example: A skilled data analyst can use business intelligence tools to not only analyze data but also come up with creative solutions to improve sales or customer service.

Development of a business intelligence system:



1. **Analysis:** This is the first step, where the team interviews workers in the organization to understand their roles. The goal is to clarify the project's objectives, priorities, costs, and benefits.
Example: A company wants to improve its sales reports, so they first talk to sales managers to understand what data they need and how the new system will help.
2. **Design:** This phase has two parts. First, the existing data system is reviewed. Then, the decision-making processes that the business intelligence (BI)

system needs to support are identified. The plan for the project is made, including phases, priorities, time, costs, and resources.

Example: If the company wants to include real-time sales data, the team checks if their current system can handle it and then plans the necessary improvements.

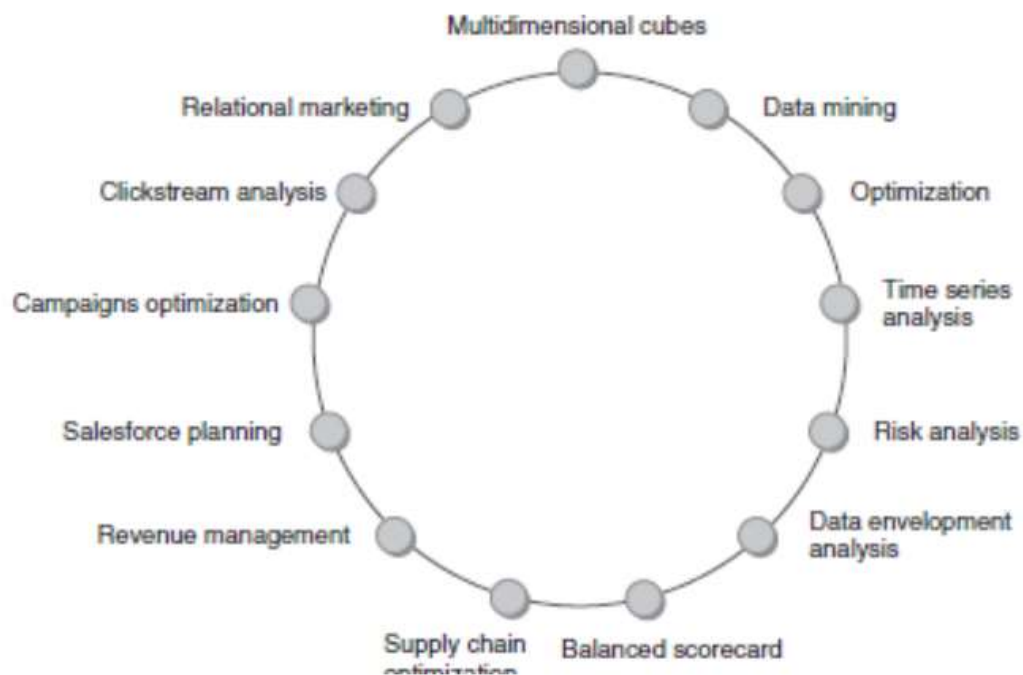
3. **Planning:** This stage involves defining the BI system's functions in detail. The data available is reviewed, and new data sources are considered. The architecture of the system, including the central data warehouse and additional data marts, is designed. The team also decides which mathematical models to use to process the data and ensures the system can handle the expected workload. A small prototype is created to check for any issues.

Example: The company plans to create a data warehouse for sales data and may add data marts for marketing and customer support. They test a small version of the system to check for issues before full development.

4. **Implementation and Control:** This final phase has five steps. First, the data warehouse and data marts are built. Then, the ETL (Extract, Transform, Load) process is set up to move data into the system. The core BI applications are developed to allow data analysis. Finally, the system is tested.

Example: The company builds its sales data warehouse and sets up automatic data transfers from the sales system. Then, they create a reporting tool for managers and test everything to ensure it works smoothly.

Each phase builds on the previous one to create a functional BI system that helps with decision-making.



Ethics and business intelligence:

1. **Data Privacy and Ethical Use:** Public and private organizations should not misuse personal data or violate people's privacy. We must prevent businesses from becoming too powerful, using these data to only benefit themselves while harming consumers, workers, and the environment.
Example: If a company uses personal data to target people unfairly or invade their privacy, it should not be allowed.
2. **Respecting Individual Rights:** Even without specific rules to stop data misuse, business intelligence analysts and decision-makers must follow ethical guidelines and respect people's privacy. The line between fair use and intrusion is especially blurred in marketing and online data analysis.
Example: A company might gather personal data under the name of 'data enrichment,' but using this data without the person's consent is not ethical, even if the name sounds harmless.
3. **Social Responsibility and Stakeholders:** Businesses should not only focus on profits but also consider the social impact of their decisions.
"Stakeholders" are anyone affected by a company's activities, including

investors, employees, unions, and society. There is debate on whether a company should focus solely on maximizing profits or also care about the community's well-being.

Example: Should a company prioritize making the most money by avoiding taxes, or should it contribute to society by paying fair taxes?

4. **Ethical Decision-Making:** Business decisions made using intelligence systems must be ethical. Analysts and decision-makers cannot stay neutral—they must consider the moral consequences of their actions.

Example: Is it ethical to design a car in a way that lowers production costs but puts passengers at risk in case of an accident?

In all these cases, businesses must think about how their use of data and decisions impact people and society, not just their profits.

UNIT -I

Business Intelligence

Decision support systems:

Definition of system, Representation of the decision-making process, Evolution of information systems, Definition of decision support system, Development of a decision support system

Decision support systems:

1. What is a Decision Support System (DSS)?

A DSS is a computer program that uses data and mathematical models to help decision-makers solve complex problems in organizations. It is especially useful for managing tasks in both public and private sectors.

Example: A company might use a DSS to analyze sales data and decide which product to focus on for the next quarter.

2. DSS in Business Intelligence:

The tools used in a business intelligence system are a type of DSS because they turn raw data into meaningful information and knowledge. This helps managers make better decisions. DSS is an essential part of building a business intelligence system.

Example: A business intelligence tool might create reports showing which

regions have the highest sales, helping managers allocate resources more effectively.

3. What This Chapter Covers:

- **Decision-Making Process Structure:** This explains how decisions are made step by step.
- **Evolution of Information Systems:** A quick overview of how information systems have developed over time.
- **Definition and Benefits of DSS:** It explains what DSSs are, their advantages, and the key factors for successfully using them.
- **Development of a DSS Project:** It describes the phases of building a DSS and the challenges in implementing it.

Example: A DSS project might involve creating a tool to help a hospital decide how to schedule doctors efficiently.

By combining data, models, and decision-making processes, a DSS simplifies complex problems and helps organizations make smarter choices.

Definition of system:

1. What is a System?

A system is a group of parts that are connected and work together to achieve a common goal.

Examples:

- **Solar system:** Planets, the sun, and moons work together in orbit.
- **Nervous system:** Brain and nerves send signals to control the body.
- **Justice system:** Laws, courts, and police work together to ensure fairness.

2. System Boundaries:

A system has boundaries that separate its inside parts (components) from the outside world.

- **Open System:** Things like materials or information can move in and out.
Example: A business gets raw materials (input), processes them, and sends out finished products (output).
- **Closed System:** Nothing goes in or out.
Example: A sealed jar of water that doesn't interact with the environment.

3. **How Systems Work:**

Every system has three main steps:

- It **receives inputs** (like materials or information).
- It **transforms** the inputs into something useful.
- It **generates outputs** (like products or results).

Example:

A factory takes raw materials (input), processes them (transformation), and creates products (output).

4. **Real-World Examples of Systems:**

- **Logistics System in a Business:**
 - Inputs: Materials, information, services.
 - Outputs: Products, deliveries, or processed information.
- **Enterprise as a System:** A company itself can be seen as a system where its boundaries, inputs (resources), and outputs (products or services) are clearly defined.

5. **Evaluating Systems (Effectiveness and Efficiency):**

- **Effectiveness:** Checks if the system achieves its goal.
Example: A sales system meeting its target for the week.
- **Efficiency:** Checks how well the system uses its resources to achieve the goal.
Example: Using fewer materials to produce more products.

6. **Feedback in Systems:**

- Feedback happens when the system's output is used as an input to improve or change its process.
Example:
- **Marketing Campaign:** A company runs ads (output), collects customer responses (feedback), and adjusts future ads based on those responses.

7. Closed Cycle Systems:

Systems that use feedback to adjust and improve their output are called closed cycle systems.

Example:

- A thermostat adjusts heating by reading the room temperature (feedback), which influences how much heat is produced.

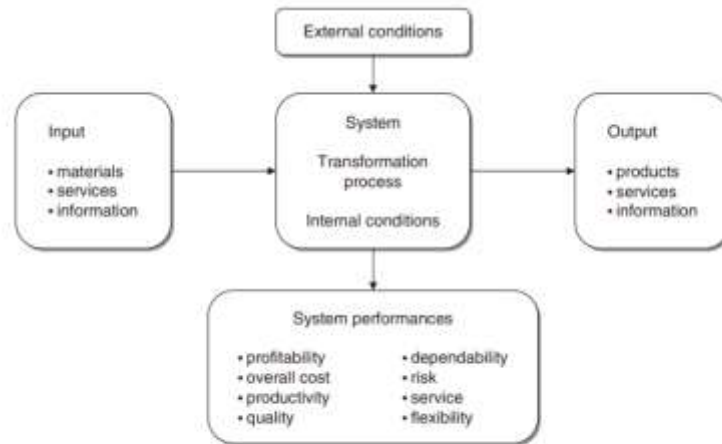


Figure 2.1 Abstract representation of a system

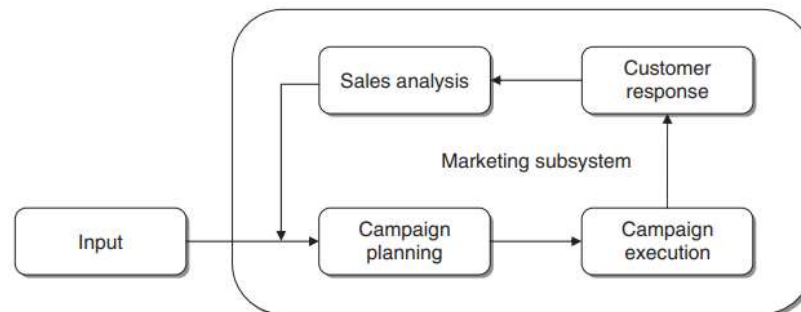


Figure 2.2 A closed cycle marketing system with feedback effects

1. Feedback in Marketing Campaigns:

- After running a marketing campaign, the sales results are collected. These results (feedback) help design the next campaign.

Example:

If a discount campaign increases sales, the company might run a similar promotion next time.

2. Why Measure System Performance?

- To evaluate how well a system is working, we measure its performance using two types of metrics: **effectiveness** and **efficiency**.

3. Effectiveness:

- Effectiveness shows how well a system meets its goals. It measures if the system is doing what it is supposed to do.
- The focus is on outputs like production levels, weekly sales, or returns on investment.

Example:

- If a factory is designed to produce 1,000 units per day, effectiveness measures whether it meets this target.

4. Efficiency:

- Efficiency measures how well the system uses its inputs (resources) to produce the desired outputs. It focuses on the quality of the transformation process.

Example:

- A company might check how much material, time, or money is needed to produce 1,000 units. Producing the same output with fewer resources means higher efficiency.

5. Key Difference Between Effectiveness and Efficiency:

- **Effectiveness:** Are we doing the *right* thing?

Example: Selling 1,000 products as planned.

- **Efficiency:** Are we doing it the *best* way?

Example: Using fewer resources to sell those 1,000 products.

Representation of the decision-making process:

To create effective Decision Support Systems (DSSs), we first need to understand how decisions are made.

- **Decision-Making Process:**

- The process involves a series of steps that guide people toward making choices or decisions.

- **Factors Influencing Decisions:**

- **Personal Attitudes:** The decision maker's own beliefs, preferences, or feelings can affect their choices.

Example: A manager might prefer eco-friendly solutions due to personal values, even if they cost more.

- **Context:** The environment or situation in which decisions are made also plays a big role.

Example: A company might make different decisions during a financial crisis compared to a period of growth.

1. **Rationality and problem solving :**

a. **What is a Decision?**

- A decision is choosing one option from several alternatives, usually done rationally.
- Everyone makes decisions daily, whether they are big or small, in both personal and professional life.

Example: Deciding whether to save money or buy something expensive.

b. **Decisions in Workplaces:**

- In workplaces, decisions are often complex and may involve:
 - **Strategic plans:** Big investment choices.
Example: A company deciding to open a new branch.
 - **Marketing plans:** Planning campaigns and predicting sales.
Example: Choosing to launch an ad campaign for a new product.
 - **Production plans:** Using resources like staff and machines effectively.
Example: Scheduling shifts to meet deadlines.

c. Decision-Making as Problem Solving:

- Decision-making is part of problem-solving. It bridges the gap between:
 - **Current state ("as is")**: Where things are now.
 - **Desired state ("to be")**: Where you want things to go.
- Moving to the desired state isn't easy and involves challenges.
- To solve problems, decision-makers:
 - a. Create alternative solutions.
 - b. Compare the pros and cons of each.
 - c. Choose the best option.

Example: If sales are low, a company might explore options like discounts, better ads, or new product launches.

d. Steps After Choosing an Option:

- Once a decision is made:
 1. It is implemented.
 2. Results are checked to see if the goal was achieved.
- If it doesn't work, the process is repeated (recursive logic).

Example: If a discount doesn't increase sales, the company might revise the strategy.

e. Alternatives in Decision-Making:

- Alternatives are possible actions to solve a problem.
- Sometimes, alternatives are few:

Example: A bank deciding whether to approve or reject a loan.
- Other times, alternatives are many or even infinite:

Example: Planning logistics for a company's annual operations.

f. Criteria for Choosing Alternatives:

- **Criteria** are measures to assess how good an option is.

- Rational decisions are based on selecting the option that meets the best criteria.
- **Economic factors** usually dominate, but there are others like technical, legal, ethical, procedural, and political factors.

g. Factors Affecting Decisions:

1. Economic Factors:

- Focus on reducing costs or increasing profits.
Example: Choosing a supplier with the lowest price.

2. Technical Feasibility:

- Discard options that are not technically possible.
Example: A factory plan that requires more capacity than available.

3. Legal Compliance:

- Ensure decisions follow the law.
Example: A marketing strategy that respects advertising regulations.

4. Ethical Principles:

- Decisions should align with community values and ethics.
Example: Avoiding child labor in supply chains.

5. Procedural Constraints:

- Some decisions may not fit an organization's culture or procedures.
Example: A company may reject a cost-effective idea if it conflicts with standard practices.

6. Political Impact:

- Consider the political consequences within or outside the organization.
Example: Deciding a department's budget cut might upset staff.

h. Decision Evaluation Process:

The process has two main stages:

• Exclusion Stage:

- Remove alternatives that don't meet basic requirements or restrictions.
Example: Discard products that exceed the project's budget.

• Evaluation Stage:

- Compare the remaining options based on performance criteria to find the best one.

Example: Choose the most reliable and cost-effective supplier from the shortlisted ones.

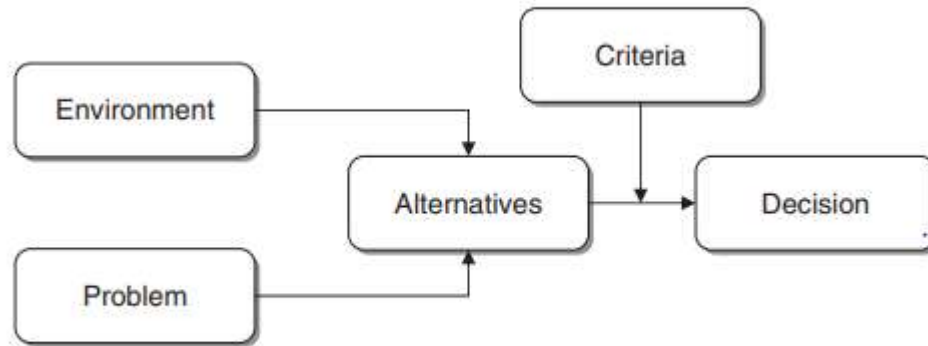


Figure 2.3 Logical flow of a problem-solving process

Decision-Making Model

A well-known model for decision-making, developed in the 1960s, is still widely used today. It consists of **five phases**: Intelligence, Design, Choice, Implementation, and Control. Here's what each phase means in simple terms:

1. Intelligence Phase

- **What happens?**
Identify and clearly define the problem by analyzing the situation and gathering information.
- **Example:**
A book delivery company notices many customer complaints about late deliveries. The issue isn't just the complaints (symptoms); the real problem might be delays in the shipping process.

2. Design Phase

- **What happens?**

Develop possible solutions or options to solve the problem. These can be explicitly listed or identified using rules and constraints.

- **Example:**

If the delivery company wants to improve shipping, possible options could include hiring more delivery staff, using faster transport methods, or partnering with local couriers.

3. Choice Phase

- **What happens?**

Evaluate all options based on specific criteria and choose the best one. Mathematical models or decision tools can help in complex situations.

- **Example:**

To decide the best shipping improvement, the company could use cost-benefit analysis or a decision tree to compare options like hiring versus partnering with couriers.

4. Implementation Phase

- **What happens?**

Turn the chosen solution into an action plan. Assign tasks, roles, and responsibilities to make it happen.

- **Example:**

If the company decides to partner with local couriers, it will sign agreements, train staff, and coordinate delivery schedules with the couriers.

5. Control Phase

- **What happens?**

After implementing the plan, check if it achieved the expected results. Compare actual outcomes to what was planned. Use this experience for future decisions.

- **Example:**

If the delivery speed improved after partnering with couriers, the company measures how many late deliveries occurred before and after the change. These results are stored for future improvements.

Key Features of the Decision-Making Process

1. Group Effort:

Decisions are often made by teams, not just one person.

- *Example:* A team of managers decides on marketing strategies, considering everyone's input.

2. Many Alternatives:

There could be countless options to choose from.

- *Example:* A company creating a yearly logistics plan has infinite ways to organize transport routes.

3. Delayed Effects:

Results of decisions usually take time to show.

- *Example:* Marketing campaigns may take months to increase sales.

4. Broad Impact:

Decisions affect multiple parts of the organization.

- *Example:* Changing the production process affects workers, costs, and product delivery.

5. Data and Analysis:

Decisions require data and a clear framework for analysis.

- *Example:* A restaurant uses customer feedback to decide which new dishes to add.

6. Feedback is Important:

Past experiences and results guide future decisions.

- *Example:* A company analyzes last year's failed marketing plan to improve this year's strategy.

7. Multiple Goals:

Decisions aim to achieve various, sometimes conflicting, goals.

- *Example:* A company might want to reduce costs but also improve product quality.

8. Uncertainty and Risk:

Many decisions involve risks or incomplete information.

- *Example:* Investing in a new technology might be risky if the market is unpredictable.

9. Cost of Trial-and-Error:

Testing decisions directly in real-life situations is often too expensive or risky.

- *Example:* Experimenting with a new factory layout without simulations could disrupt production.

10. Pressure and Speed:

Competitive environments require quick and timely decisions.

- *Example:* A tech company must release updates quickly to keep up with competitors.

2. The decision-making process

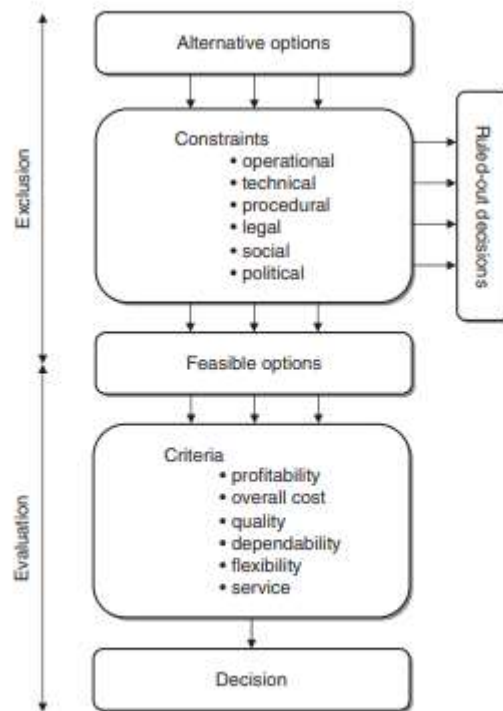


Figure 2.4 Logical structure of the decision-making process

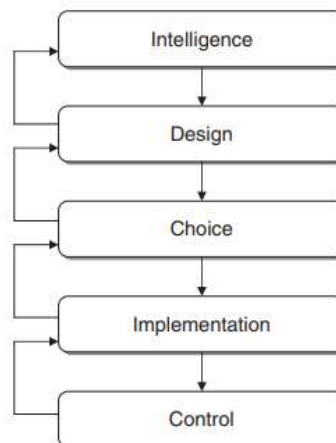


Figure 2.5 Phases of the decision-making process

The decision-making process was first explained in the 1960s and is still widely used today. It involves **five phases**:

1. **Intelligence**
2. **Design**
3. **Choice**
4. **Implementation**
5. **Control**

1. Intelligence Phase

- **What Happens:**
The decision-maker identifies and clearly defines the problem. They analyze the situation and gather all relevant information.
- **Key Point:** Don't confuse symptoms (visible issues) with the real problem.
- **Example:**
If an online bookstore receives a customer complaint about late delivery:
 - Symptom: The delivery was delayed.
 - Real Problem: The shipping department is understaffed, causing errors and delays.

2. Design Phase

- **What Happens:**
Possible solutions to the problem are created and planned. Creativity and experience play a key role here.
- **How It Works:**
 - If there are only a few solutions, list them all and pick the best.
 - If there are too many solutions, use rules or constraints to narrow down the options.
- **Example:**
For the late delivery problem:
 - Few solutions: Hire more staff or change delivery methods.

- Many solutions: Use rules like "the solution must reduce delivery time to less than 2 days."

3. Choice Phase

- **What Happens:**
Evaluate all possible solutions and select the best one using specific criteria. Mathematical models are often helpful here.
- **Tools Used:**
 - **Optimization models:** Find the best solution in complex cases with many options.
 - **Decision trees:** Handle situations with unpredictable outcomes.
- **Example:**
To fix the shipping delay, use an optimization model to calculate the cost of hiring staff versus upgrading shipping systems.

4. Implementation Phase

- **What Happens:**
The chosen solution is turned into action. Responsibilities are assigned, and the plan is executed.
- **Example:**
If the decision is to hire more staff, the implementation phase involves recruiting and training employees for the shipping department.

5. Control Phase

- **What Happens:**
The results of the implemented solution are checked against expectations. Feedback is used for future decision-making.
- **Example:**
After hiring more staff, track whether delivery times have improved and whether customer complaints have decreased. Save this data for future use.

Important Characteristics of Decision-Making

1. **Group Decisions:** Decisions are often made by a team, not one person.
 - **Example:** A project team in a company decides how to allocate resources.
2. **Many Alternatives:** There may be a large or unlimited number of options.
 - **Example:** A company choosing from hundreds of potential suppliers.
3. **Delayed Effects:** The results of decisions take time to show.
 - **Example:** Hiring new staff may take weeks to improve delivery times.
4. **Interconnected Decisions:** One decision affects multiple parts of the organization.
 - **Example:** Reducing marketing expenses may lower sales but save money.
5. **Data Access:** Decision-makers rely on data and analytical tools.
 - **Example:** A marketing team uses sales data to decide on future campaigns.
6. **Feedback:** Past experiences and results help improve future decisions.
 - **Example:** If a strategy failed previously, it's unlikely to be repeated.
7. **Multiple Goals:** Decisions often aim to achieve conflicting goals.
 - **Example:** A company wants to increase profits while reducing costs.
8. **Uncertainty and Risk:** Decisions involve risks, and people handle risk differently.
 - **Example:** Expanding into a new market could bring profits or losses.
9. **Cost of Trial and Error:** Testing decisions in the real world is too risky or expensive.
 - **Example:** A company can't test a new product launch on its entire customer base without proper research.
10. **Competitive Pressure:** Decisions must be made quickly in competitive environments.
 - **Example:** A tech company must adapt quickly to beat competitors launching similar products.

3. Types Of Decisions

Taxonomy of Decisions in DSS (Decision Support System)

Creating a **classification of decisions** (taxonomy) is helpful when designing a DSS because decisions with similar characteristics can use the same methods. Decisions are classified using **two main dimensions**:

1. **Nature of the decision**
2. **Scope of the decision**

These dimensions combine to form **nine possible categories**. This explanation focuses on the first dimension: **nature of the decision**.

Nature of Decisions

Decisions are categorized into three types based on how structured the decision-making process is:

- **Structured Decisions**
- **Unstructured Decisions**
- **Semi-structured Decisions**

1. Structured Decisions

- **Definition:**
These are decisions with a **clear process** and steps that happen repeatedly. The procedure can often be turned into an algorithm and automated.
- **Characteristics:**
 - Input, output, and processing steps are well-defined.
 - Follow predictable patterns.
 - Each phase of decision-making (intelligence, design, choice) is also structured.
- **Example:**
Approving a loan at a bank:
 - Inputs: Customer's credit score, income, and repayment history.
 - Processing: Apply specific rules (e.g., if credit score > 700 and income > \$50,000, approve the loan).
 - Output: Loan is either approved or denied.Even though this process is structured, human involvement is sometimes required for unusual situations (e.g., missing documents).

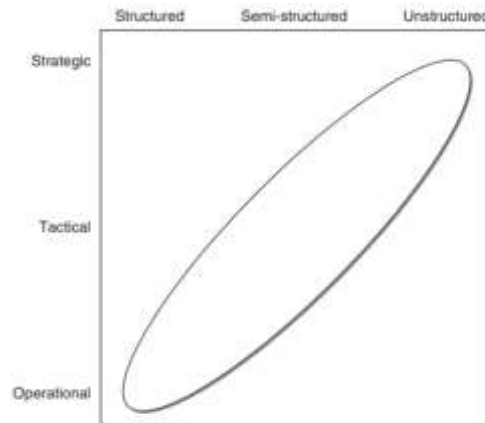


Figure 2.6 A taxonomy of decisions

2. Unstructured Decisions

- **Definition:**
A decision is **unstructured** when the three phases of decision-making (**intelligence, design, and choice**) lack clear, predefined steps.
- **Key Characteristics:**
 - At least one part of the process (inputs, outputs, or how things are processed) cannot be described in detail or turned into a fixed procedure.
 - These decisions often occur in situations that are **new or very rare**, where there is no prior experience to rely on.
- **Role of Knowledge Workers:**
People with expertise (knowledge workers) play a critical role in making these decisions because the process cannot be automated.
- **Example:** Managers analyzing market trends for a new product launch.
- **Support from Business Intelligence Systems:**
Decision-makers can use **business intelligence systems** to access **timely and flexible information**, helping them make informed choices in these uncertain situations.

Example:

Imagine a **company facing a sudden cybersecurity breach** for the first time:

1. **Intelligence Phase:** They need to identify the scope of the problem (e.g., which systems were affected).
2. **Design Phase:** They must think of possible actions (e.g., shutting down servers, hiring cybersecurity experts).

3. **Choice Phase:** They decide which action to take based on incomplete or uncertain information.

In this case, there is no predefined process to follow, and experts must analyze the situation, often relying on intuition and real-time data.

3. *Semi-Structured Decisions*

- **Definition:**

A decision is **semi-structured** when **some parts** of the decision-making process are well-organized and follow clear steps (**structured**), while **other parts** are unclear and rely on human judgment (**unstructured**).

- **Where They Occur:**

Most decisions in managing public or private organizations fall into this category.

Role of DSS and Business Intelligence

DSS (Decision Support Systems) and **business intelligence tools** can help with both structured and unstructured parts of semi-structured decisions:

1. **For Unstructured Parts:**

- **Support Type:** Passive support.
- **How It Helps:** Provides **quick and flexible access to information** to help decision-makers.
- **Example:**

A manager deciding on expanding a business might need data on market trends, competitor strategies, and customer feedback. While the tools don't give direct answers, they provide the information needed to make an informed decision.

2. **For Structured Parts:**

- **Support Type:** Active support.
- **How It Helps:** Uses **mathematical models and algorithms** to automate certain steps of the decision-making process.
- **Example:**

In the same business expansion scenario, tools can calculate potential costs, forecast profits, or identify optimal locations using algorithms.

Simplified Example:

Imagine planning a company's annual budget:

- **Unstructured Part:**

Deciding how much money to allocate for new projects requires analyzing trends, brainstorming ideas, and making judgments. Business intelligence tools provide reports and dashboards to support this decision.

- **Structured Part:**

Calculating fixed costs (e.g., employee salaries, rent) can be automated using algorithms in DSS tools.

Nature of Decisions Can Be Interpreted Differently

Sometimes, it's hard to clearly define whether a decision is structured, semi-structured, or unstructured. Different decision-makers may handle the same problem in different ways, depending on their approach and the tools they use.

Example: Setting the Sale Price of a Product

First Decision-Maker:

This person believes the best sale price can be calculated using mathematical methods, such as comparing costs and price-demand elasticity curves (how demand changes with price).

Nature of Decision: Structured (clear rules and calculations).

Example: Using a formula to calculate the price where demand is highest.

Second Decision-Maker:

This person believes price-demand elasticity doesn't consider all market factors (e.g., customer preferences or competition). Some elements can't be measured directly.

Nature of Decision: Unstructured or semi-structured (requires judgment and subjective thinking).

Example: Setting the price based on intuition, market trends, or customer feedback instead of relying only on formulas.

Decision Nature Can Vary

The same problem, such as setting a price, may seem structured to one person but unstructured or semi-structured to another, depending on their approach and the complexity of factors they consider.

Simplified and Short Version of Example 2.1: A Structured Decision

Scenario:

A paper mill cuts large standard paper sheets into smaller sizes to fulfill customer orders. Customers specify the **type, size, and quantity** of paper they need.

The Problem:

The mill must decide how to cut the sheets to **reduce paper waste**.

Why is it Structured?

This decision follows a **clear, repeatable process** supported by **optimization models** that provide the best cutting plan.

Other Industries:

This problem exists in industries like **steel, wood, glass, and fabric**, where materials are cut to size.

When Human Action is Needed:

Sometimes, **special cases** require human changes to the plan.

Example: If an important customer needs their order quickly, the company may prioritize them by adding a **fast lane**, even if it leads to more waste.

Key Point:

Structured decisions use clear rules and models but allow flexibility for special situations.

Simplified and Short Version of Example 2.2: A Semi-Structured Decision

Scenario:

A logistics manager of a company must create an **annual plan** for:

1. Allocating production volumes to plants.
 2. Purchasing materials from suppliers (volumes and delivery times).
 3. Managing production lots and stock levels.
 4. Distributing products to market areas.
-

Why is it Semi-Structured?

This decision involves **optimization models** to suggest the best plan, but:

- Some decisions depend on the **manager's judgment** instead of the model's output.
-

Example:

The optimization model might suggest buying from a cheaper supplier. However, the manager might **stick to a current supplier** (even at higher costs) because they are considered **strategic for the company's future**.

Key Point:

Semi-structured decisions mix **data-driven suggestions** with **human judgment** for balancing costs and long-term strategy.

Simplified and Short Version of Example 2.3: An Unstructured Decision

Scenario:

A company faces a **hostile takeover** attempt by a competitor through a public offer.

Why is it Unstructured?

1. The **decision-making process** depends heavily on the **specific situation and context**.
 2. It is hard to create a **step-by-step plan** that could be reused in similar cases.
-

Example:

The company may consider multiple defensive strategies, such as seeking a different buyer or legal action, but these decisions depend entirely on the unique details of the takeover situation.

Key Point:

Unstructured decisions lack a clear process and rely on **context-specific analysis and judgment**.

Classification of Decisions by Scope:**1. Strategic Decisions:**

- Impact: Affect the **entire organization** or a large part for a **long time**.
- Purpose: Set **goals and policies** for the company.
- Who Makes Them: **Top management** (e.g., CEO, board of directors).
- Example: Deciding to enter a new market or launch a new product line.

2. Tactical Decisions:

- Impact: Affect **specific parts** of the organization for a **medium-term** (up to a year).
- Purpose: Operate within the framework set by strategic decisions.
- Who Makes Them: **Middle managers** (e.g., department heads).
- Example: Planning a marketing campaign or setting production targets for a department.

3. Operational Decisions:

- Impact: Involve **day-to-day activities** with **short-term** impact.
- Purpose: Follow the guidelines from strategic and tactical decisions.
- Who Makes Them: **Lower-level managers** (e.g., supervisors, team leaders).
- Example: Scheduling employee shifts or approving small purchases.

4. Approaches to the decision-making process :**Two Main Approaches in Decision-Making:****1. Rational Approach**

- The decision maker considers factors like **economic, technical, legal, and ethical** aspects.
- The goal is to **evaluate options** and choose the best decision.
- A **Decision Support System (DSS)** helps by providing timely information and using **mathematical models**.

Example: A manager might use a DSS to analyze costs and choose the best production method.

2. **Political-Organizational Approach**

- Decisions are made more based on **instinct** and **negotiations**, without clear options or criteria.
- The DSS can only **help passively** by providing **information** and assisting during **discussions** among different managers.

Example: A decision made after several discussions among managers, based on what feels right rather than a clear set of criteria.

Two Types of Rationality:

1. **Absolute Rationality**

- The decision maker can reduce all factors into **one measurement**, like money, and choose the option that minimizes cost.
- **Example:** A manager converts all performance factors (like stock volumes and delays) into monetary values to find the cheapest solution.

2. **Bounded Rationality**

- The decision maker has limits in knowledge and resources, so they choose an option that is **good enough** based on certain thresholds.
 - **Example:** A production plan is accepted if it has **low cost, enough stock, and acceptable delivery time**.
-

Most real-world decisions are based on **bounded rationality**, where decisions are made to be **satisfactory**, not perfect.

Evolution of information systems:

What is a Decision Support System (DSS)?

- DSS combines **data** and **mathematical models** to assist decision-makers.
 - DSS is linked to an organization's **information systems** and evolved alongside them.
-

Timeline of Information Systems Evolution

1. **1940s–1960s: Early Data Processing**
 - Digital computers automated routine tasks in areas like **administration** and **production** to improve efficiency.

2. **1970s: Management Information Systems (MIS)**
 - Companies developed MIS to provide **timely and useful information** for decision-makers.
 - Challenges:
 - Limited technology: Mainframe computers lacked **graphics** and relied on **text-based terminals**.
 - Centralized IT departments caused **delays** in implementing changes.

3. **Late 1980s: Personal Computers (PCs)**

- PCs with **graphic interfaces** and tools like **spreadsheets** made workers more **independent** of centralized IT systems.
- Workers created **local databases** and **models**, paving the way for modern business intelligence tools.

Example: Spreadsheets helped managers simulate scenarios and make decisions without waiting for IT support.

4. **1980s–1990s: Executive and Strategic Information Systems**

- Systems for **executives** supported **unstructured decisions** by providing easy access to information.
 - These were **passive tools**, focused on timely information delivery.
-

5. **1990s: Client-Server Models and Data Warehouses**

- Network-based **client-server models** became popular.
 - Databases for DSS were separated from operational systems, leading to concepts like **data warehouses** and **data marts**.
-

6. **Late 1990s: Business Intelligence (BI)**

- The term **BI** was introduced to describe systems combining DSS, **analytics**, and **data transformation** into knowledge for decision-makers.
-

Summary:

DSS and BI evolved from basic data automation to advanced tools that empower decision-makers with **useful information** and **independent processing capabilities**. Modern BI integrates data, models, and tools to make better decisions.

Definition of decision support system :

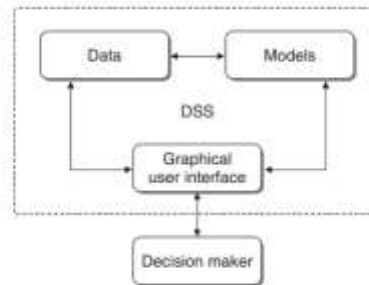


Figure 2.8 Structure of a decision support system

1. What is a Decision Support System (DSS)?

A DSS is a computer system that helps decision-makers solve semi-structured and unstructured problems by combining **data**, **mathematical models**, and **user interaction tools**.

- Example: Think of a DSS as a tool like Google Maps. It uses data (maps), models (shortest path algorithms), and allows users to interact (input destination) to find the best route.

2. Key Features of DSS:

- **Effectiveness:**
A DSS helps make better decisions by offering deeper analysis, though it may take more time.
 - Example: A marketing team uses a DSS to analyze customer data before launching a campaign, leading to better targeting.
- **Mathematical Models:**
Uses mathematical tools like statistics and operations research to turn data into insights, unlike simple systems that only provide data.
 - Example: A company uses a profit prediction model to decide the best product to launch.
- **Integration in Decision-Making:**
A DSS helps in individual and group decisions for all types of processes—strategic, tactical, or operational.
 - Example: A retail store uses a DSS to decide weekly stock levels by analyzing past sales.
- **Organizational Role:**
DSS encourages communication between different levels and tracks decision processes.
 - Example: A DSS tracks decisions made by a sales team and shares them with management.
- **Flexibility:**
A DSS adapts to changes in user needs or the environment and has user-friendly interfaces, often via web browsers.

- Example: A company adjusts its DSS to include new tax laws for financial planning.
-

3. Components of a DSS (Structure):

- **Data Management:**
A database stores all the data needed for decision-making. It is often connected to a data warehouse.
 - Example: A company's sales data stored in a DSS helps managers forecast revenue.
 - **Model Management:**
Provides mathematical tools like optimization and statistical models for analysis.
 - Example: A logistics company uses optimization models in a DSS to plan deliveries efficiently.
 - **Interactions:**
Users input data and get results through simple, graphical interfaces.
 - Example: A manager uses a web interface to input sales goals and gets a bar chart of potential outcomes.
 - **Knowledge Management:**
Connects to the company's knowledge base to incorporate shared expertise.
 - Example: A DSS suggests solutions based on past decisions stored in the system.
-

4. Advantages of a DSS:

- **More Options:** Users can analyze more alternatives.
 - Example: Evaluating 5 marketing strategies instead of just 2.
- **Better Decisions:** Leads to higher-quality decisions.
 - Example: Choosing a product to discontinue based on profit trends.
- **Deeper Understanding:** Helps users understand problems better.
 - Example: Analyzing market demand patterns for better product placement.
- **Scenario Analysis:** Tests "what-if" scenarios by changing assumptions.
 - Example: Simulating profit changes if prices increase by 10%.
- **Faster Reactions:** Allows quick responses to unexpected events.
 - Example: Adjusting inventory during a sudden demand spike.
- **Data Utilization:** Extracts maximum value from available data.
 - Example: Using customer data for personalized recommendations.
- **Improved Communication:** Enhances collaboration among teams.
 - Example: Sharing sales forecasts with marketing and supply chain teams.
- **Better Teamwork:** Facilitates group decisions effectively.
 - Example: Teams use DSS to evaluate projects during brainstorming sessions.

- **Reliable Controls:** Makes decision processes clearer and more trustworthy.
 - Example: Auditing past financial decisions using DSS logs.

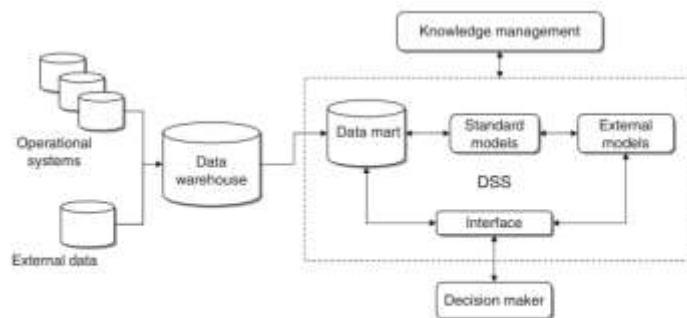


Figure 2.9 Extended structure of a decision support system

Development of a decision support system :

Development Phases of a Decision Support System (DSS):

1. **DSSs are Unique:**
 - Unlike common software, DSSs are custom-built based on specific needs.
 - Example: You can't buy a ready-made DSS like you can buy a word processor.
2. **Phases in DSS Development:**
 - The process involves **feedback loops**, meaning adjustments might be needed at any stage.
 - Example: Fixing errors after testing a prototype.

1. Planning:

- **Purpose:** Understand needs and decide if the DSS should be developed.
- Includes a feasibility study:
 - Questions like "Why do we need this DSS?"
 - Determine benefits, costs, and timeline.
- Example: A DSS for sales forecasting helps improve decision quality, not just speed.

2. Analysis:

- **Purpose:** Define what the DSS should do and who will use it.

- Study decision processes, current data, and future needs.
 - Example: Mapping current sales decisions and planning how the DSS will improve them.
-

3. Design:

- **Purpose:** Define how the DSS will work.
 - Choose hardware, software, databases, and user interfaces.
 - Example: Using a web browser for easy user access to reports and data.
-

4. Implementation:

- Build, test, and install the DSS.
 - Address organizational impacts using change management.
 - Example: Train employees to use the DSS so they don't feel left out.
-

Rapid Prototyping:

- Develop small, testable parts of the DSS step by step.
 - Benefits: Errors can be corrected early, and user feedback improves the system.
 - Example: Build a module for data input first, then add analysis features.
-

Agile and Extreme Programming:

- Speed up DSS development by frequently updating and testing.
 - Example: Regularly involve users to refine the tool.
-

Ongoing Updates:

- DSSs need regular changes as situations evolve.
 - Example: Update the DSS when a new product line is launched.
-

Critical Success Factors:

1. **Integration:**

- Requires different tools, models, and teams to work together.
- Example: A system integrator ensures hardware, software, and processes align.

2. **Involvement:**

- Decision-makers and users must participate in development.
- Example: Include sales managers in planning so they trust the system.

3. **Uncertainty Management:**

- Reduce risks through prototypes, tests, and user-friendly designs.
- Example: Test a prototype on a small team before full deployment.

By following these steps and ensuring proper collaboration and testing, a DSS can improve decision-making effectively.